9. April, Friday 18. More Proof Techniques (3.1, 3.2). Order Axioms

Technique (Proving Universal Statements)

To prove a statement of the form "for all $a \in A$, we have $P(a)$"

We fix an arbitrary element $a \in A$ and prove $P(a)$ holds

Can only use $a \in A$.

Example: For all integers x, if x is even then x^2 is even.

NOT a proof: Let 8 be an even integer. Then $8^2 = 64 = 2 \cdot 32$ is even.

Theorem 1: For any real number a, we have $0 + a = a$.

Proof: Fix a real number a. By the "existence of identities" axiom, we know that $a + 0 = a$. By commutativity, we also know that

$$0 + a = a + 0.$$

Since $0 + a = a + 0$ and $a + 0 = a$, by the Transitivity axiom we get $0 + a = a$, as desired. □

Theorem 2: In the real numbers, the additive identity is unique.
(There is only one real number with the property $a+0=a$ for every $a \in \mathbb{R}$)

Technique: (Proving uniqueness) In order to prove something is unique,
1) suppose you have two of that thing, say x and y
2) Show $x = y$.

Proof of Theorem 2: Suppose $0' \in \mathbb{R}$ is another additive identity, so $b+0' = b$ for any $b \in \mathbb{R}$

Since $0'$ is an additive identity, we get $0+0' = 0$.

Since 0 is an additive identity, we get $0'+0 = 0'$.

But by commutativity, $0+0' = 0'+0$.

So by substitution $0' = 0$, as desired □

Another proof: Assume $a+0=a$ and $a+0'=a$ for any $a \in \mathbb{R}$.
By symmetry, we have $a = a+0'$. Since $a+0=a$ and $a = a+0'$
by transitivity we have $a+0 = a+0'$. By cancellation, (8.2) we get $0=0'$ □

Theorem 3

a) For every $a \in \mathbb{R}$, we have $0 \cdot a = 0$.

b) For every $a \in \mathbb{R}$, we have $(-1) \cdot a = -a$

c) Additive inverses are unique.

Proof: HW. Can't use the results we are going to proof today. □

Theorem 4: $(-1) \cdot (-1) = 1$

Proof: Since -1 is the additive inverse of 1, we have $1 + (-1) = 0$.

Then, by Theorem 3.1 we obtain

$$\big(1 + (-1)\big) \cdot (-1) = 0 \cdot (-1)$$

By Theorem 3(a) we obtain

$$\big(1 + (-1)\big) \cdot (1) = 0$$

And by distributivity, we have

$$1 \cdot (-1) + (-1)(-1) = 0$$

Using that 1 is a multiplicative identity, we have $1 \cdot (-1) = -1$, so

$$-1 + (-1)(-1) = 0$$

And by commutativity, we have

$$(-1)(-1) + (-1) = 0$$

On the other hand, we know that $1 + (-1) = 0$, so $0 = 1 + (-1)$ by symmetry

Thus, $(-1)(-1) + (-1) = 0$ and $0 = 1 + (-1)$

By Transitivity we get $(-1)(-1) + (-1) = 1 + (-1)$

And by cancellation law (Thm 7.2), we get $(-1)(-1) = 1$. □

Order Axioms for $\mathbb{R}$

We assume there is a subset $\mathbb{R}^+ \subseteq \mathbb{R}$ with the following properties:

12) Positive closure: If $a, b \in \mathbb{R}^+$, then $a+b \in \mathbb{R}^+$ and $ab \in \mathbb{R}^+$.

13) Trichotomy: For any $a \in \mathbb{R}$, exactly one of the following is true:

1) $a \in \mathbb{R}^+$, 2) $-a \in \mathbb{R}^+$, or 3) $a = 0$.

Elements in $\mathbb{R}^+$ are called positive. We denote $\mathbb{Z}^+ = \mathbb{Z} \cap \mathbb{R}^+$ and $\mathbb{Q}^+ = \mathbb{Q} \cap \mathbb{R}^+$

Definition 5: For real numbers a, b,

- $a < b$ means that $b - a \in \mathbb{R}^+$
- $a \leq b$ (leq) means $a < b$ or $a = b$
- $a > b$ means $a - b$ is positive (i.e. $b < a$)
- $a \geq b$ (geq) means $a > b$ or $a = b$
- a is negative means $-a$ is positive.

Theorem 6: Let $a \in \mathbb{R}$. Then $a \in \mathbb{R}^+$ if and only if $a > 0$.

Technique: To prove a biconditional $P \longleftrightarrow Q$, prove $P \longrightarrow Q$ and $Q \longrightarrow P$.

Lemma 7: We have $-0 = 0$.

Proof: Since 0 is the additive identity, $0 + 0 = 0$.
But this equation also says that 0 is an additive inverse of 0.
By Theorem 2(c), additive inverses are unique, so $0 = -0$. □

Proof of Theorem 6. Fix $a \in \mathbb{R}$.

Need -0. So let's write a lemma.

[$\Rightarrow$] (\Rightarrow) Suppose $a \in \mathbb{R}^+$. We want to show that $a > 0$, in other words, that $a - 0 \in \mathbb{R}^+$.

So we compute:

$$a - 0 = a + (-0) = a + 0 = a$$

(Notation; Lemma 6; Additive id.)

Since we know that $a \in \mathbb{R}^+$ and $a - 0 = a$, we also get $a - 0 \in \mathbb{R}^+$ by substitution so $a > 0$ as desired.

[$\Leftarrow$] Suppose now $a > 0$, and we want to show that $a \in \mathbb{R}^+$. (\Leftarrow)

By definition, $a > 0$ means $a - 0 \in \mathbb{R}^+$. But again,

$$a - 0 = a + (-0) = a + 0 = a$$

Since $a - 0 \in \mathbb{R}^+$ and $a = a - 0$, we get $a \in \mathbb{R}^+$, as desired. $\square$

Theorem 8: 1 is positive.

Technique (Proof by contradiction) Assume the Theorem is False. Deduce a contradiction. Therefore your assumption is wrong, so the Theorem is True.

Proof of Theorem 8: Suppose for a contradiction that 1 is not positive.

By Trichotomy, either -1 is positive or $1=0$.

Case 1: -1 is positive. By closure of $\mathbb{R}^+$ we get that $(-1)(-1) \in \mathbb{R}^+$

But by Theorem 3, $(-1)(-1) = 1$, so we get $1 \in \mathbb{R}^+$, This is a contradiction!

Case 2: $1=0$. This immediately contradicts axiom 5, which states that $1 \neq 0$.

Either case gives a contradiction, so 1 must be positive $\square$